UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	COMPUTER ORGANIZATION AND ARCHITECTURE
COURSE CODE	CSC429
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of two (2) parts: PART A (10 Questions) PART B (1 Question)
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 4 printed pages.

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PART A (80 MARKS)

Answer ALL questions in this part.

QUESTION 1 (5 marks)

a) Describe THREE (3) types of system interconnection and its function.

b) Differentiate the physical appearance between system interconnection and CPU interconnection.

QUESTION 2 (5 marks)

Fill in the blank with the most relevant answer.

a) In the
Generation, the needs of the air conditioning is not compulsory.

b) The acronym EDSAC stands for
Generation.

c) The Ultra Large Scale Integrator is one of the features in the

d) is known as the father of modern computer.

e) Gottfried von Liebniz is the inventor for
mechanical machine.

QUESTION 3 (5 marks)

a) Calculate 408 - 40h using sign and magnitude. Show your complete workflow.

b) Calculate 39h - 23a using 1 's and 2's complement method. Show your complete workflow.

QUESTION 4 (5 marks)

Interaction between computer and external environment can be facilitated by using input and output devices. Without these input / output devices, a computer is just a dumb machine that is not able to operate effectively and efficiently to perform a specific task.

a) State ONE (1) device that can be both input and output device.

b) Briefly describe ONE (1) scenario when your answer in (a) can be input and ONE (1) scenario that it can be output.

c) Sketch a basic Von Neumann computer architecture with appropriate label.

QUESTION 5 (10 marks)

a) Discuss the main function of cache memory.

b) How can the data stored in the cache and main memory be synchronized.

c) Contrast between SRAM and DRAM based on the following characteristics:

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Type of storage
ii,
Availability of charges leak
price
Example of location of memory usage

QUESTION 6 (10 marks)

a) List the FOUR (4) classifications of parallel processors and provide examples for each.

b) Discuss ONE (1) advantage of using cluster computer architecture.

QUESTION 7 (10 marks)

Describe the specificfunction ofdifferent types of registers.

QUESTION 8 (10 marks)

a) Calculate FIFAie - 664i6 using the following techniques. Showthe complete workflow.

   i. Sign and magnitude

   ii. 2's complement

QUESTION 9 (10 marks)

a) Discuss TWO (2) types of bus timing.

b) Describe THREE (3) Central Processing Unit (CPU) components.

QUESTION 10 (10 marks)

a) Discuss THREE (3) main key factors for memory selection.

b) List TWO (2) examples for internal cabling memory and TWO (2) examples of system memory.

PART B (20 MARKS)

Answer the question in this part.

QUESTION 11 (20 marks)

Question 1 is based on Figure 1 and Table 1.
Write an appropriate program using correct assembly language notations to compute the equation shown in Figure 1 using instructions for one-address, two-address, and three-address. The available instructions are provided in Table 1. The register labeled A, B, C, and D are fixed, while registers x, y, and z can be used temporarily. The final answer must be stored in register y.

$$y = A - \frac{2C}{B + A^2} * D$$

Figure 1

Table 1 : Machine instruction

One-address	Two-address	Three-address
LOADp	MOV (p,q)	ADD (p.q.r)
STORp	ADD (p,q)	SUB (p,q,r)
ADDp	SUB (p,q)	MUL (p.q.r)
SUBp	MUL (p,q)	DIV(p,q,r)
MULp	DIV(p,q)	
DIVp		